GLP-1 · THE MOLECULES

Eight molecules.
Two have arrived.

GLP-1 medicines copy the gut hormones that tell the body it has eaten. Two are approved for obesity: semaglutide and tirzepatide. Six more are in trials.

587%

growth in US prescriptions for these medicines between 2019 and 2024.

What follows: what each molecule targets, how far it has got, and which diseases it is aimed at.

No trade names, companies or share prices. Sources on the final slide.

HOW THEY WORK

One target became four.

	GLP-1	GIP	amylin	glucagon
semaglutide	●	○	○	○
tirzepatide	●	●	○	○
cagrilintide + semaglutide	●	○	●	○
aleniglipron	●	○	○	○
pemvidutide	●	○	○	●
zenagamtide	●	○	●	○
petrelintide	○	○	●	○
ACCG-2671	○	○	●	○

The first medicines hit one receptor. The newer ones pair GLP-1 with a second — to add effects, not just dose harder.

A filled dot means the molecule targets that receptor.

WHERE EACH ONE STANDS

Two on the market.
Six behind them.

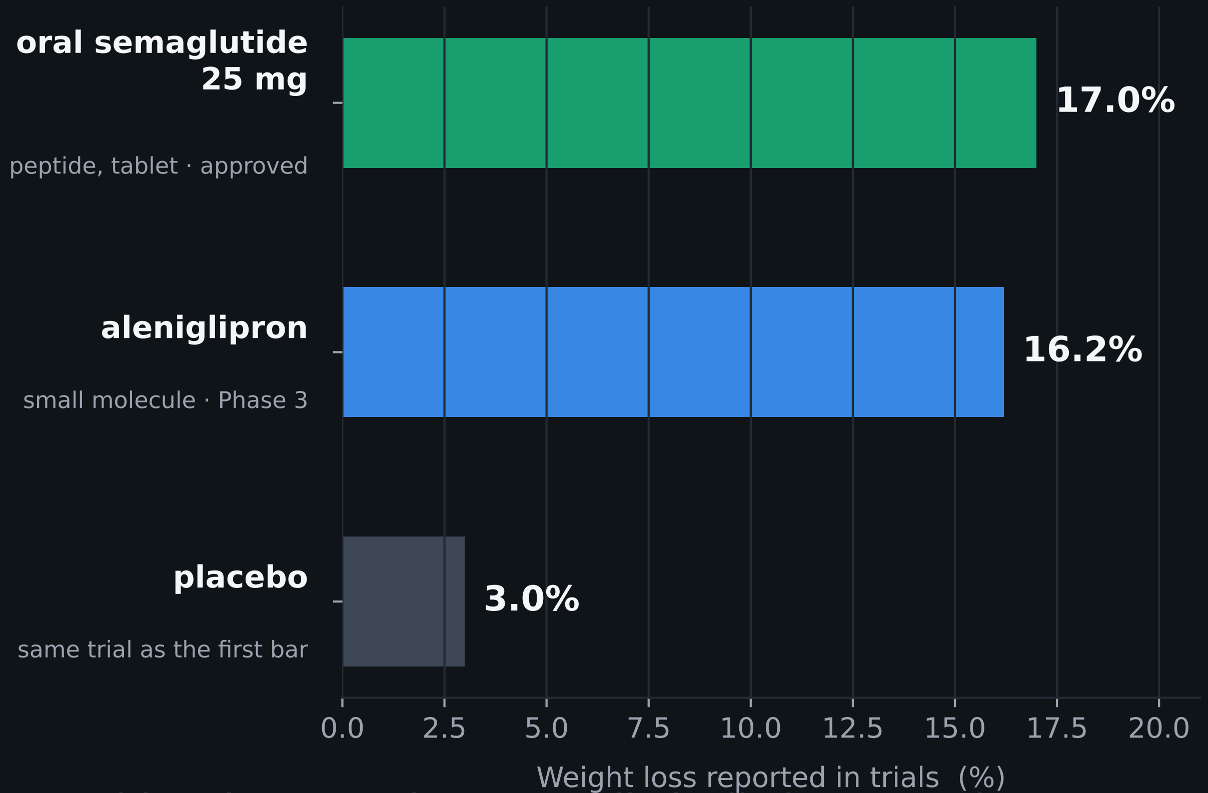
Phase 1	Phase 2	Phase 3	Filed	Approved
ACCG-2671 amylin	**petrelintide** amylin	**aleniglipron** GLP-1	**cagrilintide + semaglutide** GLP-1 · amylin	**semaglutide** GLP-1
		pemvidutide GLP-1 · glucagon		**tirzepatide** GLP-1 · GIP
		zenagamtide GLP-1 · amylin		

Only two mechanisms have reached patients. Everything else — oral formats, combinations, new receptors — is still being tested.

Stage as at August 2026. Development stages change; several readouts are due.

THE SHIFT TO A TABLET

The next contest is oral.



A tablet changes who can realistically be treated, not just how much weight comes off. One is a peptide in tablet form; the other a small molecule, simpler to manufacture at scale.

Not a head-to-head comparison. Different trials, populations and durations: 17% versus 3% placebo at Phase 3b; 16.2% is the upper end of an open-label extension, with a 10.4% discontinuation rate.

BEYOND OBESITY

The same biology is moving into the liver.

Obesity / overweight	Liver disease (MASH)	Alcohol use disorder
semaglutide	semaglutide	pemvidutide
tirzepatide	pemvidutide	
cagrilintide + semaglutide		
aleniglipron		
zenagamtide		
petrelintide		
ACCG-2671		

Adding glucagon acts directly on the liver while GLP-1 handles appetite. That widened the field from weight to liver disease, and now to addiction.

One molecule is approved in MASH, one of only two treatments for it. A glucagon/GLP-1 agonist began Phase 3 there in Aug 2026; data due 2029.

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SCOPE AND SOURCES

What this covers.

Receptor targets, approval status and trial stage as at August 2026, from company filings, company releases and industry landscape reviews.

No trade names. No share prices, valuations or company comparisons — this is a view of the science, not of the market.

Trial results are point-in-time and stages change. Cross-trial figures are not head-to-head.

Sources

SEC filings (10-Q, 8-K) and company releases, Dec 2025 – Aug 2026; regulatory announcements; industry landscape reviews.

Related work

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